

LIFE CONTINGENCIES I - ACTU 301

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June 25, 2025

SURVIVAL MODELS
AGE-AT-DEATH RANDOM VARIABLE

Abstract

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- The central difficulty in issuing *Life insurance* is that of determining *the time of death of the insured*
- Let time O denote the time of birth of an individual. We will assume everyone is alive at birth.
- Let x be a continuous and non negative random variable, Then x is an approximate random variable to model the age at which an individual dies and is therefore called *Age at Death*

Definition 1 : The cumulative distribution of x is the probability that a new born dies before age x and is given by:

$$F_o(x) = p(X \leq x) = p(X < x)$$

$F_o(x) \equiv$ The probability that death will occur prior to or at age x

The Distribution function satisfy three properties :

- $F_o(0) = 0$
- $\lim_{x \rightarrow \infty} F_o(x) = 1$